

potentiomap: An R package for reproducible potentiometric surface mapping

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Summary

potentiomap is an R package for preparing groundwater observations, interpolating potentiometric surfaces, exporting map products, and deriving hydraulic-gradient flow arrows. The package supports common hydrogeologic input forms: direct groundwater elevation measurements, depth-to-water measurements paired with a DEM, depth-to-water measurements paired with land-surface elevation columns, and depth-to-water measurements paired with separate surface-elevation points. Outputs include **terra** rasters, contour vector layers, quicklook graphics, hydraulic-gradient rasters, sampled gradient points, flow-arrow line layers, and arrow-tip or arrow-base point layers.

The package is intended for applied hydrogeologists, environmental scientists, and geospatial analysts who need reproducible potentiometric mapping workflows without repeatedly rebuilding project-specific scripts. It provides a smooth thin-plate spline surface as the default interpolation product while retaining support for inverse distance weighting, ordinary kriging, universal kriging, and user-supplied interpolation functions.

Statement of Need

Potentiometric surface maps are a standard product in groundwater monitoring, remediation, water-resource assessment, and compliance reporting. In practice, these products are often produced through a mixture of spreadsheet handling, desktop GIS operations, and project-specific scripts. That workflow can make it difficult to document assumptions, compare interpolation methods, rerun maps as new measurements arrive, or reproduce hydraulic-gradient arrows consistently.

potentiomap addresses this need by putting the complete workflow in a small R package built on established spatial tools. Raster operations and vector exports use **terra** (Hijmans 2025), thin-plate splines use **fields** (Nychka et al. 2021), and geostatistical interpolation uses **gstat** (Pebesma 2004). The package standardizes groundwater elevation observations into a simple point layer with **Z** and **Name** fields, then carries that structure through interpolation, contour development, visual review, and hydraulic-gradient products.

Functionality

The main functions are organized around the mapping workflow:

- **ps_make_points()** standardizes direct groundwater elevation measurements.
- **ps_potentiometric_points()** calculates groundwater elevation from depth-to-water measurements and land-surface elevation information.
- **ps_interpolate()** creates one or more surface rasters. The default method is thin-plate spline interpolation (TPS); IDW, OK, UK, and custom interpolation functions are also supported.
- **ps_smooth_surface()** applies optional focal smoothing to an interpolated raster before contour or arrow development.
- **ps_contours()** and **ps_export_surfaces()** create contour and GIS deliverables.
- **ps_flow_arrows()** derives hydraulic gradients and downgradient arrow lines.
- **ps_arrow_vertices()** extracts arrow-tip or arrow-base point layers.

The following example uses the bundled synthetic monitoring data:

```
library(potentiomap)
library(terra)

data("synthetic_wells")
data("synthetic_dem")
synthetic_dem <- terra::rast(synthetic_dem)

gw_points <- ps_potentiometric_points(
  synthetic_wells,
  x = "x",
  y = "y",
  depth_col = "depth_to_water",
  surface = synthetic_dem,
  name_col = "well_id",
  crs = "EPSG:26916"
)

surfaces <- ps_interpolate(
  gw_points,
  methods = c("TPS", "IDW", "OK", "UK"),
  grid_res = 50,
  mask = ps_sample_aoi()
)

smoothed_tps <- ps_smooth_surface(surfaces$TPS, window_size = 5)
flow <- ps_flow_arrows(smoothed_tps, res_factor = 8, scale = 80)
tips <- ps_arrow_vertices(flow$arrows, which = "last")
```

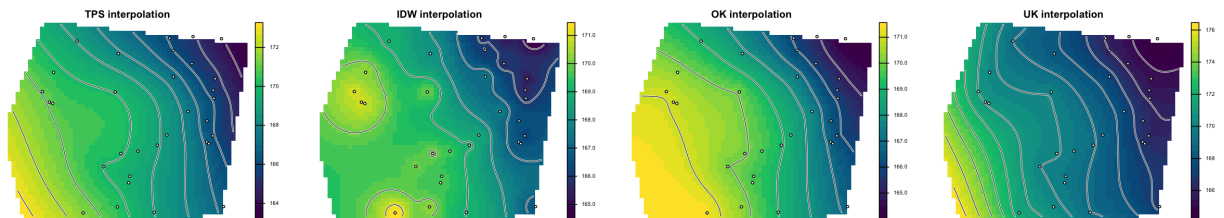


Figure 1: Examples of interpolation methods available in `potentiomap`.

Availability and Quality Control

`potentiomap` includes synthetic example data so documentation, tests, and examples can be run without project-specific datasets. The test suite exercises direct water-level input, depth-to-water conversion, all built-in interpolation methods, custom interpolation functions, raster smoothing, contour generation, surface export, quicklook PNG creation, hydraulic-gradient raster export, arrow line export, and arrow-tip/base extraction. The repository also includes developer scripts outside the R package source tree for local example generation and CRAN-style checking.

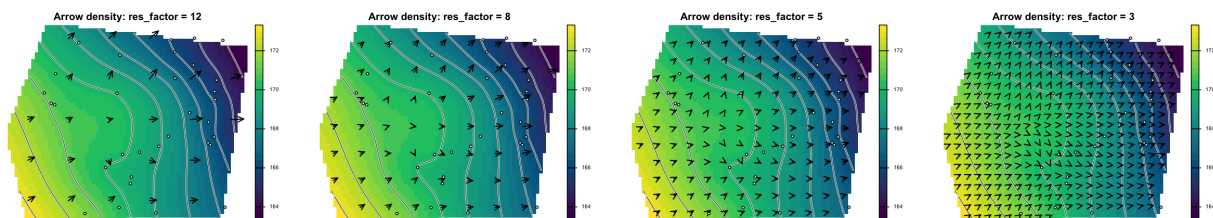


Figure 2: Examples of arrow-density controls available in `potentiomap`.

Acknowledgements

This package builds on the R geospatial ecosystem, especially `terra`, `fields`, and `gstat`.

References

- Hijmans, Robert J. 2025. *terra: Spatial Data Analysis*. <https://doi.org/10.32614/CRAN.package.terra>.
- Nychka, Douglas, Reinhard Furrer, John Paige, and Stephan Sain. 2021. *fields: Tools for Spatial Data*. <https://doi.org/10.5065/D6W957CT>.
- Pebesma, Edzer J. 2004. “Multivariable Geostatistics in S: The gstat Package.” *Computers & Geosciences* 30: 683–91. <https://doi.org/10.1016/j.cageo.2004.03.012>.